AC9M6SP03 • Year 6 Mathematics

Transformations and Tessellations

Combine translations, rotations and reflections without gaps or overlaps

We are learning to…

Students analyse how repeated congruent shapes cover a plane, describe the transformation sequence and use angle relationships to test whether vertices fit around a point.

Represent | Reason | Calculate | Interpret

Verify

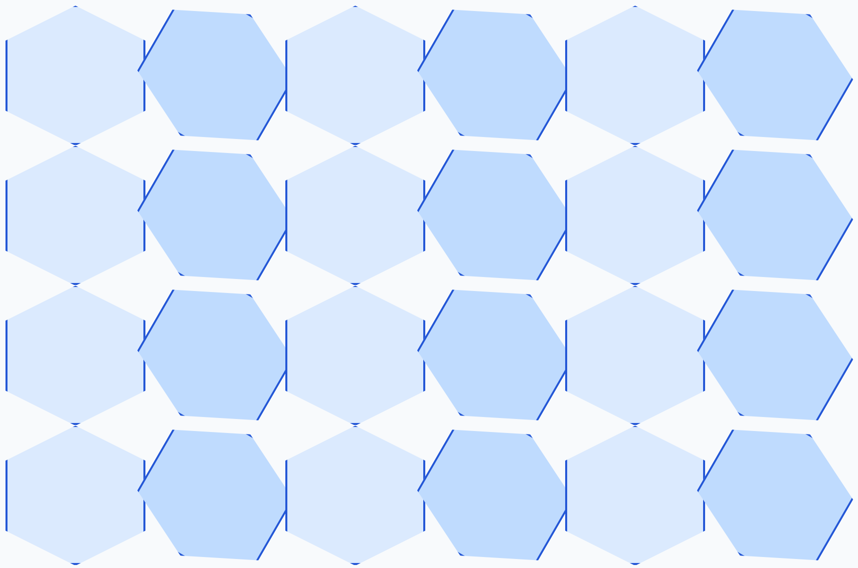
Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

recognise and use combinations of transformations to create tessellations and patterns, using digital tools where appropriate

Build a repeating tessellation



A repeated shape covers the plane without gaps or overlaps

A tessellation preserves congruence under rigid transformations. Around each vertex, the meeting angles total 360°.

Use transformations to explain the pattern

move	description
translation	**slide motif by a fixed vector**
rotation	**turn motif about a vertex**
reflection	**flip motif across a line**
glide reflection	reflect, then translate along the line
vertex test	meeting angles total 360°

Digital tools can repeat precise transformations, but students should identify the generating motif and transformation rule.

Build and annotate the model

Represent transformations and tessellations and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Pattern with gaps called tessellation:** The plane must be covered continuously.
- **Shape resized during repetition:** Rigid transformations preserve congruence.
- **Transformation described vaguely as move:** Name direction, distance, centre, angle or line.
- **Vertex angle condition ignored:** Use 360° to test feasibility.

Quick check

- Identify a tessellation.
- Name the motif.
- Describe a translation.
- Use a vertex-angle test.
- Combine transformations.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.